

40 LinkedList Split a Circular linked list into two halves.

```
class Solution {
public:
    pair<Node *, Node *> splitList(struct Node *head) {
        if (!head) return {nullptr, nullptr};

        // If only one node, both halves will point to head (circular)
        if (head->next == head)
            return {head, head};
        Node *slow = head;
        Node *fast = head;
        // Move `fast` by 2 and `slow` by 1 until fast reaches end of list
        while (fast->next != head && fast->next->next != head) {
            slow = slow->next;
            fast = fast->next->next;
        }
        // For even number of nodes, move fast one more step to point to last node
        if (fast->next->next == head)
            fast = fast->next;
        // Now slow is at end of first half, fast is at last node
        Node *head1 = head;
        Node *head2 = slow->next;
        // Make the two halves circular
        slow->next = head1;
        fast->next = head2;
        return {head1, head2};
    }
};
```

The screenshot displays a coding platform interface with a dark theme. On the left, the 'Output Window' shows 'Compilation Results' for a submission by 'Y.O.G.I. (AI Bot)'. It indicates 'Problem Solved Successfully' with a green checkmark. A summary table shows 'Test Cases Passed' as 1115 / 1115, 'Attempts: Correct / Total' as 1 / 1, 'Accuracy' as 100%, 'Points Scored' as 2 / 2, and 'Time Taken' as 0.15. Below this, 'Your Total Score' is 138. At the bottom, 'Solve Next' buttons are visible for 'Insert in Sorted Circular Linked List', 'Middle of a Linked List', and 'Split Linked List Alternatingly'. On the right, the code editor shows the C++ solution for the problem, with line numbers 1 through 35. The code is identical to the one provided in the previous block.

Test Cases Passed	Attempts: Correct / Total
1115 / 1115	1 / 1

Accuracy: 100%

Points Scored	Time Taken
2 / 2	0.15

Your Total Score: 138

Solve Next

- Insert in Sorted Circular Linked List
- Middle of a Linked List
- Split Linked List Alternatingly